# Budget Bites Project Report

AjxOm undergraduate Project One
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Budget Bites is a device-local lunch notebook. Students save meal estimates, record actual purchase amounts, correct receipt history and compare a transparent history-based estimate with a lunch budget. The application works without an account or a network service.

## Implemented scope

| Screen role | Delivered behavior |
| --- | --- |
| Catalog | Saved meal cards, search/cuisine/favorites filters, add/edit/delete and budget comparison |
| Meal editor | Validated meal name, venue note, item labels and USD prices |
| Detail and activity | Start, draft progress, explicit save/finish, over-budget confirmation, reset, local helper |
| History | A.J. card/list layout integrated with receipt CRUD, date/meal filters and spending total |
| Settings | Budget slider and dark mode stored on-device |

## Product and data choices

Two SQLite entities, Meal and PurchaseLog, keep catalog estimates separate from historical actual purchases. Favorites is a flag on Meal. Prices use integer cents. The starter meals and QA purchase fixtures are fictional. No current restaurant prices, delivery, payment, remote inference or cloud database are implied.

The approved proposal and signed statement accompany the project package.

# Architecture and integration

A.J. authored the initial Meal model, SQLite methods and standalone History page on the data branch. Om's integration branch preserved those commits, corrected incompatible types/navigation and connected the final History screen to PurchaseLog. Both histories are combined through a merge rather than rewritten or reassigned.

## Class responsibilities

| Component | Responsibility |
| --- | --- |
| BudgetBitesApp | App theme and lifetime of AppStore/database |
| HomeScreen | Tabs, meal actions and navigation callbacks |
| MealHistory | Receipt list, meal/date filtering, total and Card/ListTile presentation |
| MealEditor and PurchaseEditor | Validated transient forms and pending-save state |
| AppStore | Shared meals/purchases, refresh and preferences |
| MealDatabase | Connection owner, constraints, migration and parameterized CRUD |
| Domain helpers | Exact money parsing and pure Lunch Lens calculation |

## Integrity and lifecycle

PurchaseLog.mealId references Meal.id with restricted deletion. Linked receipts must be removed or reassigned before the parent can be deleted. Receipt deletion requires confirmation. A stable unique draft token makes repeat/retry saves idempotent. Updates refresh both totals and estimates. Editing the catalog estimate never rewrites a prior actual amount.

Version 2 converts the earlier REAL dollar fields into integer cents while preserving IDs. A device fixture verifies that ID 17 survives with a 650-cent total. Forms dispose controllers and guard asynchronous UI updates after navigation. Reset clears only the unsaved draft. SharedPreferences stores budget and dark mode; active-draft process restoration is outside scope.

## Integration problems resolved

The original table had a trailing SQL comma and REAL/int price conflicts. The standalone History page indexed an integer ID as a string and invoked a private State class instead of navigating. Integration keeps the shared database, uses valid initials and routes receipt edits to PurchaseEditor. History deletes child receipts, not catalog meals.

# Local helper and evaluation

Lunch Lens is a local rule-based moving-average estimate. It uses the three latest purchases of the selected meal strictly before the chosen prediction time. It shows the inputs, date range, estimate and last-price baseline. Not now dismisses it. It never silently fills the actual amount or saves a purchase.

| Case | Expected and actual result | Baseline / reference |
|---|---|---|
| H1 normal: 450,600,750 cents | 600 cents; pass | 750 / 650 |
| H2: only 450,600 | Insufficient-history feedback; pass | Not scored |
| H3: 450,-1,750 | Invalid-data exception; pass | Not scored |
| H4: 100,100,101 | 100 cents; pass | 101 / 100 |
| H5: 500,2000,500 | 1000 cents; pass | 500 / 500 |

These are designed fictional development fixtures, not real spending or held-out evidence. Across the three valid comparisons, mean absolute error is 183.33 cents for the mean and 33.67 cents for last price, n=3. Empty and invalid cases assess feedback rather than numeric prediction accuracy.

## Design rationale and evaluation

Meal identity, amount and timestamp are relevant inputs. Sensitive personal information is unnecessary. Three recent records are easy to explain but can mix promotions and extras. A median-of-three is a reasonable alternative. The outlier case demonstrates that the baseline can win. We retain the brief's moving mean with the limitation disclosed and user control preserved, without claiming broad accuracy or causal benefit.

The implementation excludes future/reference rows, uses deterministic timestamp/ID ties, rounds to integer cents and recomputes after corrections. AI_FEATURE.md records rationale, alternative, cases, metrics, counterexample and user-control/offline evidence.

# Testing and contributions

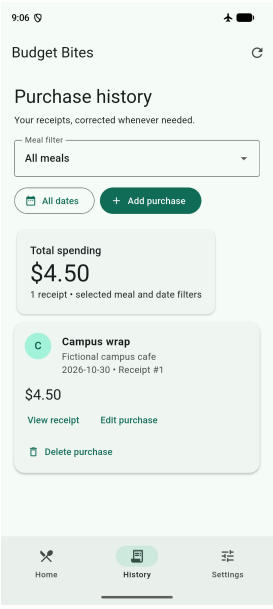
| Check | Recorded result |
| --- | --- |
| Analyzer and unit/widget tests | Final logs identify the exact integration result. Suite includes 17 logic/widget checks, including three new History checks. |
| Two emulator profiles | API 37 Pixel_8 and a separate API 37 emulator with a larger display. Actual SQLite CRUD, duplicate-token, FK and migration tests. |
| Release and restart | Release APK built. The integrated release was built, installed and checked after offline force-stop/relaunch on both emulator profiles. |
| Responsive behavior | Narrow History with enlarged text has a targeted automated check. Earlier landscape/large-text screenshots remain dated development evidence. |

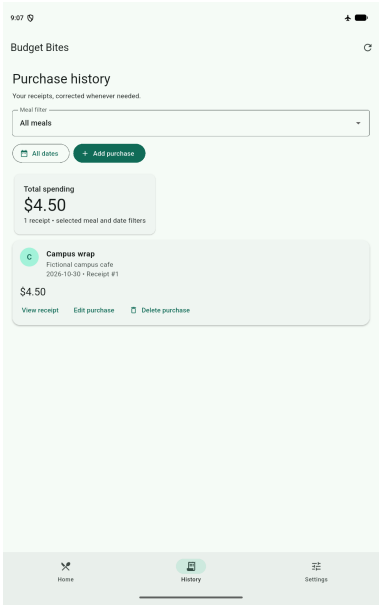
Final validation logs: evidence/final-analyze.txt, final-unit-widget-tests.txt, final-phone-integration.txt, final-large-integration.txt and final-release-build.txt. The October 26 logs document the earlier development checks.

## Team contributions

A.J. owns the original data foundation and History-page contribution. Om is responsible for frontend integration, navigation, corrected persistence wiring, helper, tests and packaging.

## Release validation captures




Actual installed-release History views from the two emulator profiles. Displayed records are fictional test fixtures.

Course references: codd.cs.gsu.edu/~lhenry23/mad/Projects/Project1/ver2/assessment/grading-rubric.html and submission/About_Submission.html. Repository: github.com/Omswaroopreddyudumula/ajxom-budget-bites.